

Intro to ML — Milestone 1 Report

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1 Introduction

Predicting addictive behaviors from behavioral data is a canonical supervised learning task. We study the Gaming and Mental Health dataset (2000 synthetic samples, 13 gaming-habit features) across two tasks: regression of a continuous addiction score, and 3-class classification of addiction level (Low, Medium, High). We compare three method families: linear regression (Linear Regression, Kernel Ridge), linear classification (Logistic Regression, Linear SVM), and instance-based prediction (k -Nearest Neighbors). Severe class imbalance between majority and minority classes makes accuracy misleading and drives macro-F1 as our primary classification metric; we also report accuracy and MSE. Hyperparameters are selected by 5-fold cross-validation.

2 Methodology

2.1 Data & Preprocessing

The 13 features span more than three orders of magnitude in standard deviation (largest ≈ 514 , smallest ≈ 0.19), so we z-score normalize using training-set statistics. Class counts: Low 68.9%, Medium 27.9%, High 3.2%; a majority-class predictor reaches 70.0% accuracy but only 0.275 macro-F1. The regression target is right-skewed (mean 2.79, median 2, range 0–10). We take a stratified 80/20 split of the training set as our held-out test, preserving the 3.2% High class in both halves.

2.2 Cross-Validation

Folds are stratified for classification ($n=51$ High samples in train would otherwise risk empty validation folds), and the final model is refit on all training data before evaluation. We cross-checked our 5-fold estimates against 10-fold CV on LinReg; the two agreed within fold std across the λ sweep, while a single held-out split was noticeably noisier. Large search spaces (LogReg’s $\eta \times \beta \times T$; KNN’s $k \times \text{metric} \times \text{weighted}$) use coarse-to-fine refinement; the resulting sweep curves are summarized in Fig. 1.

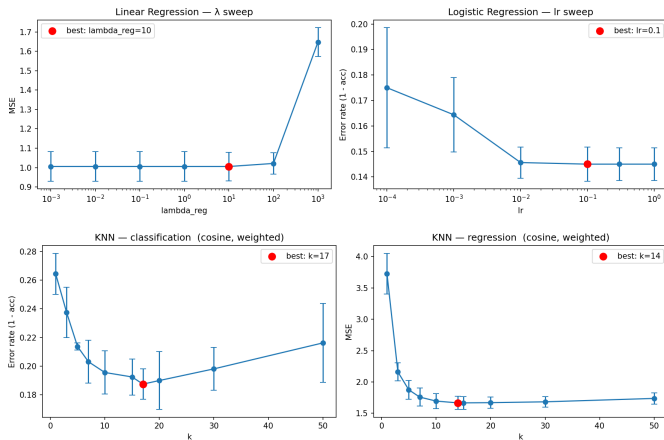


Figure 1: 5-fold CV hyperparameter sweeps for the required methods. (a) LinReg ridge λ (log- x , MSE); (b) LogReg learning rate (log- x , 1-acc); (c,d) KNN neighbor count k for classification (1-acc) and regression (MSE). Red markers = selected value.

3 Methods & Results

3.1 Linear regression for addiction score (LinReg, KRR)

We fit linear regression by three solvers of the same objective $\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}_{\setminus b}\|^2$: (i) closed-form ridge $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ (bias unregularized), (ii) gradient descent, and (iii) Kernel Ridge Regression (KRR) via the dual form $\boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1} (\mathbf{y} - \bar{y})$ with linear, polynomial or RBF kernels. The parametric path also supports polynomial feature expansion (per-feature or with cross-terms).

The three solvers coincide on the linear problem: CF and GD converge to identical weights ($\|\Delta \mathbf{w}\|/\|\mathbf{w}\| \approx 10^{-16}$), and KRR with the *linear* kernel recovers the ridge solution as expected. Adding non-linearity hurts: polynomial cross-terms ($d=2+\text{int}$) overfit (train 0.92, CV 1.08, Fig. 2b); $d=3$ lies within the 0.068 fold std of baseline; KRR with RBF $\sigma=1$ reaches CV MSE 3.68, polynomial $d=3$ reaches 2.61, and only wide- σ RBF ($\sigma \geq 10$) approaches the linear kernel. Three independent parametrizations converge on the same conclusion: the regression signal is near-linear in the 13 z-scored features.

We select the simpler ridge solution: CF ridge, $\lambda=10$ (Fig. 1a), $d=1$, test MSE **0.993**; KRR with the linear kernel matches it at 0.969, within fold noise (Table 1).

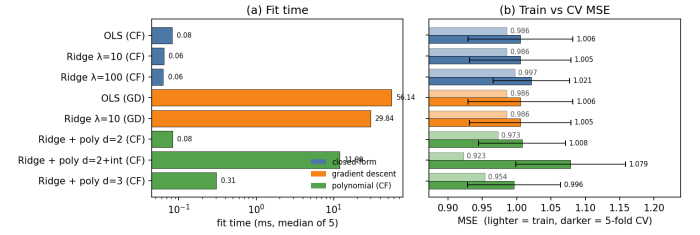


Figure 2: Linear regression variants. (a) Median fit time (log-scale). (b) Train MSE (light) vs 5-fold CV MSE (dark, $\pm$ std). Selected: CF ridge, $\lambda=10$, $d=1$.

3.2 Linear classification for addiction level (LogReg, SVM)

Both Logistic Regression and Linear SVM parametrize a linear decision boundary $\mathbf{W}^\top \mathbf{x}$; they differ only in the surrogate loss—multinomial softmax cross-entropy vs multi-class hinge—so a head-to-head isolates the effect of the loss function. LogReg supports Xavier init, momentum, early stopping, L2 regularization (bias excluded), inverse-frequency class-weighted CE, polynomial expansion, and mini-batch SGD. SVM shares the same optimizer options, making the two directly comparable.

At $\eta=0.3$ (inside the flat region in Fig. 1b) with 1000 iters, every LogReg optimizer enhancement converges to macro-F1 ≈ 0.758 in 5-fold stratified CV (Fig. 3b): momentum, Xavier init, and early stopping are redundant once the optimizer has enough steps. Mini-batch SGD at $bs=32$ matches this F1 at $\sim 4\times$ less fit time (30 ms vs 130 ms); pure SGD is too noisy (F1 0.655).

The one knob that improves the primary metric is *inverse-frequency class weighting* (F1 0.769, -1.4 pp accuracy)—a trade the $21\times$ imbalance makes worth taking. Over-regularization collapses minority F1 ($\lambda=100$: 0.513); polynomial cross-terms overfit ($d=2+\text{int}$: 0.726). Switching to hinge loss (SVM, $\lambda=10^{-2}$) gives CV F1 0.764—within noise of LogReg—but class weighting *hurts* SVM (0.746); we did not fully diagnose why. The Kernel Least-Squares

Classifier (KLS, one-hot multi-output KRR) with polynomial $d=2$ reaches CV F1 0.723, still below either parametric linear classifier.

Final models: class-weighted LogReg (test F1 0.706) and SVM $\lambda=10^{-2}$ (test F1 **0.733**); the margin loss edges the probabilistic one by ~ 0.03 on held-out (Table 1).

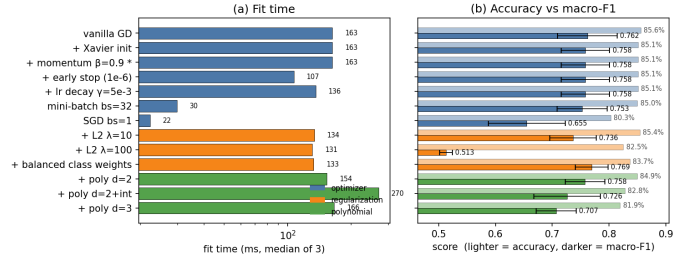


Figure 3: Logistic Regression variants. (a) Median fit time (log scale). (b) 5-fold stratified CV accuracy (lighter) and macro-F1 (darker, $\pm$ std). Selected: class-weighted CE with momentum.

3.3 Instance-based prediction (KNN, both tasks)

We implemented a unified KNN for both tasks: majority vote for classification and mean target for regression, with L1, L2 or cosine distance and optional inverse-distance weighting of the neighbors.

A full grid over $k \times \text{metric} \times \text{weighted}$ (5-fold stratified CV for classification, random 5-fold for regression) picks the same winning configuration for both tasks: $k=17$, cosine, inverse-distance weighted (Fig. 4). Cosine dominates the other two metrics by a clear margin on both panels—after z-score normalization the features already live at comparable scales, so directional (cosine) similarity carries more signal than absolute L1/L2 distance. Small k overfits (classification F1 drops to ≈ 0.58 at $k=1$); large k under-fits; the curves bottom out in the $k \in [10, 20]$ range. Inverse-distance weighting consistently outperforms uniform weighting; predict time is metric-dependent and linear in N (see §4.1). Selected model ($k=17$, cosine, weighted) achieves 80.1% test accuracy and macro-F1 **0.623** for classification, and test MSE **1.62** for regression (Table 1).

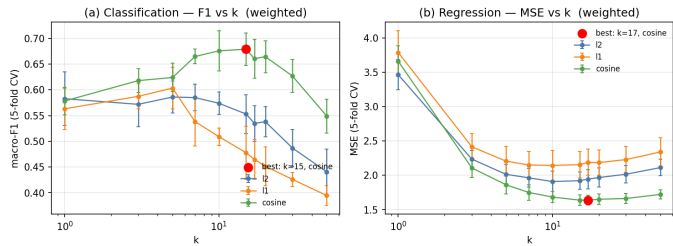


Figure 4: KNN across k with three distance metrics (inverse-distance weighted). (a) 5-fold stratified CV macro-F1, classification. (b) 5-fold CV MSE, regression. Selected: $k=17$, cosine, weighted.

4 Discussion & Conclusion

4.1 Latency and scalability

Fit and predict costs split the methods sharply. Closed-form LinReg fits in ~ 0.1 ms ($\sim 500\times$ faster than GD). LogReg and SVM mini-batch take 30–60 ms. KNN’s predict is $O(N)$ per query (37–238 ms for $N=1600$, metric-dependent), scaling linearly in training-set size—ruling it out for latency-sensitive deployment. Kernel methods sit

at the opposite extreme: the $N \times N$ solve is $O(N^3)$ in fit and $O(N)$ in predict, with $O(N^2)$ memory. For this dataset ($N=1600$) all methods run comfortably, but the scaling gap would dominate at $N \gtrsim 10^4$: only the parametric linear models (LinReg, LogReg, SVM) would remain practical at that scale.

4.2 Best models per task

Table 1 consolidates the selected configuration and held-out test performance for every method; all three required methods meet or exceed the handout’s sanity-check baselines. **Regression** is essentially a linear problem: ridge LinReg (test MSE 0.993) matches KRR’s linear kernel (0.969) within fold-to-fold noise, so we report the simpler solution; KNN trails at 1.62. **Classification** is tighter: SVM ($\lambda=10^{-2}$) achieves the best test macro-F1 (0.733), a ~ 0.03 margin over class-weighted LogReg (0.706); KLS (0.619) and KNN (0.623) trail by ~ 0.1 F1, driven by poor recall on the 11-sample High test minority (Fig. 5). Low- and Medium-class F1 are near-tied between SVM and LogReg (± 0.01); the margin comes from the High class, where both recover 7/11 samples but LogReg’s inverse-frequency weighting over-commits to the minority (7 High false positives vs SVM’s 4) and drops its High-F1 to 0.56 (SVM: 0.64)—a 3-sample swing on the 11-sample minority, where each prediction moves macro-F1 by ~ 0.02 , so the 0.03 gap itself should not be over-interpreted. Both parametric classifiers still beat the non-parametric and kernel alternatives. Across all five methods the story is consistent: the signal is near-linear in the 13 z-scored features, so parametric linear models win both tasks by small margins, with the remaining bottleneck being minority-class recall—a data problem more than a model problem. **What didn’t work:** polynomial/RBF expansions, over-regularization (LogReg F1 collapses to 0.513 at $\lambda=100$), very small k (KNN F1 0.58 at $k=1$), and pure SGD (F1 0.655).

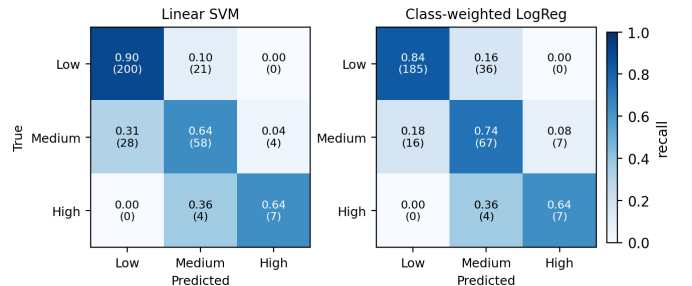


Figure 5: Test-set confusion matrices (row-normalized recall) for the two best classifiers. Both capture Low near-perfectly but struggle on the 11-sample High class.

Table 1: Selected configurations and test-set performance. Reg: MSE (lower = better); Clf: accuracy, macro-F1 (higher = better).

Method	Task	Config	Test
LinReg	reg	CF ridge $\lambda=10$	MSE 0.993
KRR	reg	linear kernel, $\lambda=1$	MSE 0.969
KNN	reg	$k=17$, cos, wtd	MSE 1.62
LogReg	clf	bal. weights, $\eta=0.3$, $\beta=0.9$	80.4% / F1 0.706
SVM	clf	$\lambda=10^{-2}$, $bs=32$	82.3% / F1 0.733
KLS	clf	poly $d=2$, $\lambda=1$	81.7% / F1 0.619
KNN	clf	$k=17$, cos, wtd	80.1% / F1 0.623